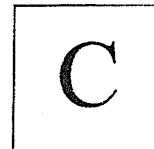


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B.Tech. Degree VI Semester Special Supplementary Examination November 2022

ME 19-205-0604 HEAT AND MASS TRANSFER

(2019 Scheme)

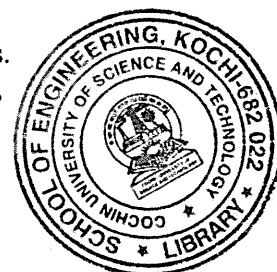
Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basic modes of heat transfer.
 CO2: Get an insight on conduction, convection and radiation heat transfers.
 CO3: Apply the knowledge of heat transfer through various geometries for optimizing the thickness of insulation.
 CO4: Understand the unsteady heat conduction and its applications.
 CO5: Get a concept on multiphase flow, diffusion and convective mass transfer.
 CO6: Attain information of parallel and counter flow heat exchangers and their design aspects.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome



PART A (Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	Write short notes on critical thickness of Insulation.	3	L3	3	1
(b)	Show that the temperature profile for the heat conduction through a wall of constant thermal conductivity is a straight line.	3	L2	2	1,2
(c)	Write short notes on Ficks law of diffusion.	3	L1	5	1,2,3
(d)	Compare film wise and drop wise condensation.	3	L2	1	1,2,3
(e)	State and prove Kirchhoff's identity.	3	L2	4	1,2,3
(f)	Write short notes on Radiation shields.	3	L2	4	1,2,3
(g)	Explain NTU method of heat exchanger analysis.	3	L3	6	1,2,3
(h)	Write short notes on Classification of Heat exchanger.	3	L1	6	1,2,3

PART B

(4 × 12 = 48)

- II. A hollow cylinder with inner radius 30 mm and outer radius 50 mm is heated at the inner surface at a rate of 105 W/m^2 and dissipated heat by convection from outer surface into a fluid at 80°C with heat transfer coefficient of $400 \text{ W/m}^2\cdot\text{K}$. There is no energy generation and thermal conductivity of the material is constant at $15 \text{ W/m}\cdot\text{K}$. Calculate the temperatures of inside and outside surfaces of the cylinder.
- OR**
- III. A person is found dead at 5 p.m. in a room where temperature is 20°C . The temperature of the body is measured to be 25°C when found, and the heat transfer coefficient is estimated to be $8 \text{ W/m}^2\cdot\text{K}$. Modelling the human body a 30 cm diameter, 1.70 m long cylinder, calculate actual time of death of the person. Take thermo physical properties of the body:

$$k = 6.08 \text{ W/m}\cdot\text{K}, \rho = 900 \text{ kg/m}^3, C = 4000 \text{ J/kg}\cdot\text{K}.$$

(P.T.O.)

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		Marks	BL	CO	PO
IV.	Air at 10°C and at a pressure of 100 kPa is flowing over a plate at a velocity of 3 m/s. If the plate is 30 cm wide and at a temperature of 60°C. Calculate the following quantities at $x = 0.3$ m. (i) Boundary layer thickness (ii) Local friction coefficient (iii) Local shearing stress (iv) Total drag force (v) Thermal boundary layer thickness (vi) Local convective heat transfer coefficient (vii) The heat transfer from the plate, if the length of the plate is 0.5 m.	12	L4	5	1,2,3
OR					
V.	Consider a rectangular plate 0.2 m × 0.4 m is maintained at a uniform temperature of 80°C. It is placed in atmospheric air at 24°C. Compare the heat transfer rates from the plate for the cases when the vertical height is (i) 0.2 m (ii) 0.4 m.	12	L4	5	1,2,3
VI.	A plane, gray, diffuse and opaque surface (absorptivity = 0.7) with a surface area of 0.5 m ² , is maintained at 500°C and receives radiant energy at a rate of 10,000 W/m ² . Determine per unit time: (i) The energy absorbed (ii) The radiant energy emitted (iii) The total energy leaving the surface per unit area (iv) The radiant energy emitted by the surface in the wave band 0.2 μm to 4 μm (v) The net radiative heat transfer from the surface.	12	L3	2	1,2,3
OR					
VII.	Two concentric cylinders have inner and outer radii 5 cm and 10 cm, respectively and length 20 cm. Calculate the all possible view factors.	12	L3	4	1,2,3
VIII.	A heat exchanger is required to cool 55,000 kg/h of alcohol from 66°C to 40°C using 40,000 kg/h of water entering at 5°C. Calculate: (i) exit temperature of water (ii) heat transfer rate (iii) surface area required for: (1) parallel flow type (2) counter flow type of heat exchanger. Take overall heat transfer coefficient $U = 580$ W/m ² .K, C_p (alcohol) = 3760 J/kg.K C_p (water) = 4180 J/kg.K.	12	L5	6	1,2,3
OR					
IX.	A chemical having specific heat of 3.3 kJ/kg.K at a rate of 20,000 kg/h enters a parallel flow heat exchanger at 120°C. The flow rate of cooling water is 50,000 kg/h with an inlet temperature of 20°C. The heat transfer area is 10 m ² and overall heat transfer coefficient is 1050 W/m ² .K. Find: (i) The effectiveness of the heat exchanger (ii) Outlet temperature of water and chemical. Take C_p of water as 4.186 kJ/kg.K	12	L5	6	1,2,3

Bloom's Taxonomy Levels

L1 = 8.33%, L2 = 16.6%, L3 = 58.3%, L4 = 33.33%, L5 = 50%.
